

Degrees of Freedom Estimation in the Meta-analysis of Sensitivity and Specificity in Diagnosis Medicine

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Abstract

When we use Wald F test to test the statistical significance of the fixed effect in the joint mixed effect model with binary outcomes, an approximate method is needed to estimate the denominator degrees of freedom of the Wald F statistic. In this study, we examine and compare the small sample type I error rate control of three different approximation methods via simulation. Data simulation is conducted based on a meta-analysis of sensitivity and specificity in the diagnosis medicine. Our result shows that Kenward-Roger method and Between-Within method have a better small sample performance than the normal approximation method. However, under our settings, all the three approximation methods would inflate type I error rate when the total cluster number is small.

1 Introduction

Hypertension is a serious medical condition that significantly increases the risks of heart, brain, kidney and other diseases. According to the World Health Organization, an estimated 1.13 billion people worldwide have hypertension in 2019.[1] The diagnosis of hypertension relies on measurement of a patient's blood pressure (BP). However, accurate measurement of BP is challenging, particularly in busy office practices where medical assistants often measure BP before the patient is seen by the clinician. The oscillometric office BP measurement is the most convenient, however, not the most accurate method of diagnosis. Home BP monitoring (HBPM) and 24-hour ambulatory BP monitoring (ABPM) are out-of-office alternatives, with the latter considered as the reference standard for BP measurement.

To systematically review the accuracy of the conventional oscillometric office BP measurement method and HBPM for correctly classifying adults as having hypertension and to examine whether there is any difference between the office BP measurement and HBPM, Dr. Feng-Chang Lin and his Duke colleagues

conducted a meta-analysis of sensitivity and specificity in diagnosis medicine. In their study, a total of 12 cross-sectional studies (N=6877) comparing conventional oscillometric office BP to ABPM average and 6 studies (N=2049) comparing HBPM to ABPM average were included, with a range of 117 to 2209 participants per analysis. The counts of true positives (comparator technique also showed BP elevation), true negatives (comparator technique also showed no BP elevation), false positives (comparator technique showed BP elevation while ABPM did not), and false negatives (comparator technique showed no BP elevation while ABPM showed BP elevation) were recorded for each study. Sensitivity (True Positive Rate) measures the proportion of positives that are correctly identified. Specificity (True Negative Rate) measures the proportion of negatives that are correctly identified. The sensitivity and specificity were calculated in **Table 1** for each study, respectively. Then a meta-analysis of sensitivity and specificity follows.

Table 1: Number of True Positives, Number of False Negatives, Number of True Negatives, Number of False Negatives, Sensitivity and Specificity of Office oscillometric BP measurement and HBPM compared to ABPM for each study

First Author, Year	Screening Test	Total N(hypertension %)	True Positives	False Negatives	True Negatives	False Positives	sensitivity	specificity
Gill	Office BP	211 (41)	23	12	63	113	0.27 (0.18-0.37)	0.90 (0.84-0.95)
Ishikawa	Office BP	129 (34)	31	11	13	74	0.70 (0.57-0.84)	0.87 (0.80-0.94)
Kang	Office BP	573 (64)	266	75	102	130	0.72 (0.68-0.77)	0.63 (0.57-0.70)
Kim	Office BP	121 (54)	34	3	31	53	0.52 (0.40-0.64)	0.95 (0.89-1.00)
Ommen	Office BP	178 (25)	24	39	20	95	0.55 (0.40-0.69)	0.71 (0.63-0.79)
Poudel	Office BP	432 (38)	40	8	122	262	0.25 (0.18-0.31)	0.97 (0.95-0.99)
Selenta	Office BP	319 (27)	12	1	73	233	0.14 (0.07-0.22)	1.00 (0.99-1.00)
Shin	Office BP	1262 (61)	554	188	222	298	0.71 (0.68-0.75)	0.61 (0.57-0.66)
Tocci	Office BP	2209 (67)	1332	351	149	377	0.90 (0.88-0.91)	0.52 (0.48-0.55)
Viera	Office BP	411 (76)	109	5	205	92	0.35 (0.29-0.40)	0.95 (0.90-0.99)
Wojciechowska	Office BP	201 (33)	41	20	25	115	0.62 (0.50-0.74)	0.85 (0.79-0.91)
Zhang	Office BP	831 (62)	197	26	321	287	0.38 (0.34-0.42)	0.92 (0.89-0.95)
Gill	HBPM	211 (41)	62	30	24	95	0.72 (0.61-0.81)	0.76 (0.68-0.83)
Ishikawa	HBPM	120 (33)	22	13	18	67	0.55 (0.40-0.70)	0.84 (0.76-0.92)
Kang	HBPM	573 (64)	247	55	121	150	0.67 (0.62-0.72)	0.73 (0.67-0.79)
Nunan	HBPM	203 (54)	102	46	7	48	0.94 (0.89-0.98)	0.51 (0.41-0.61)
Zhang	HBPM	831 (62)	295	33	223	280	0.57 (0.53-0.61)	0.89 (0.86-0.93)
Zhuo	HBPM	102 (71)	63	8	9	22	0.88 (0.80-0.95)	0.73 (0.57-0.89)

2 Methods

In the meta-analysis of sensitivity and specificity, A joint logistic regression model with random intercept was fitted to calculate the random effects bivariate summary measures. The joint logistic regression model is the extension of logistic regression model by joining the likelihood functions of two logistic regression models together. Let N denote the number of studies, n_{1i} denotes the sum of true positives and false negatives in study i , and n_{2i} denotes the sum of true negatives and false positives in study i , where $i = 1, 2, \dots, N$ is the index of each study. The joint logistic regression model can be expressed as follows:

$$\log\left(\frac{p_{1i}}{1-p_{1i}}\right) = \alpha_0 + x_i\alpha_1 + b_{1i} \quad (1)$$

$$\log\left(\frac{p_{2i}}{1-p_{2i}}\right) = \beta_0 + x_i\beta_1 + b_{2i} \quad (2)$$

$$\begin{pmatrix} b_{1i} \\ b_{2i} \end{pmatrix} \sim N(0, \Sigma), \quad (3)$$

where p_{1i} and p_{2i} denote the sensitivity and specificity of study i , respectively. α_0 and β_0 are scalars, which denote the intercepts in the logistic regression model. α_1 and β_1 are scalars, which denote the fixed-effects logistic regression parameters. x_i is a scalar, which denotes the indicator of home BP measurement method for study i . $x_i = 1$ equals 1 if study i compares HBPM to ABPM, and equals 0 if study i compares office BP measurement to ABPM. b_{1i} and b_{2i} are scalars, which denote the random intercepts for study i in the mixed effect logistic regression model of sensitivity and specificity, respectively. $b_i = (b_{1i}, b_{2i})^T$, where b_i follows a bivariate normal distribution with mean 0 and variance-covariance matrix Σ , where

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}.$$

Accordingly, parameter $\theta = (\alpha^T, \beta^T, \sigma_1^2, \sigma_2^2, \rho)^T$, where $\alpha = (\alpha_0, \alpha_1^T)^T$ and $\beta = (\beta_0, \beta_1^T)^T$.

The parameters in the joint mixed effects logistic regression model can be calculated by using either the standard maximum likelihood estimation or the restricted maximum likelihood estimation. We will use the standard maximum likelihood estimation in this project. In the logistic regression model of sensitivity, for each study, we have an indicator of home BP measurement method, x_i , the number of true positives y_{1i} , and the number of false negatives $n_{1i} - y_{1i}$. The sensitivity (True Positive Rate) is denoted as p_{1i} and the False Negative Rate is denoted as $1 - p_{1i}$. Similarly, in the logistic regression model of specificity, for each study, we have an indicator of home BP measurement method, x_i , the number of true negatives y_{2i} , and the number of false positives $n_{2i} - y_{2i}$. The specificity (True Negative Rate) is denoted as p_{2i} and False Positive Rate is denoted as $1 - p_{2i}$. Then the likelihood can be expressed as

$$L(\theta|Y) \propto \prod_{i=1}^N p_{1i}^{y_{1i}} (1 - p_{1i})^{n_{1i} - y_{1i}} p_{2i}^{y_{2i}} (1 - p_{2i})^{n_{2i} - y_{2i}} \quad (4)$$

The log-likelihood turns products into sums,

$$\ell(\theta|Y) \propto \sum_{i=1}^N \ell_i(\theta|y_i), \quad (5)$$

where

$$\ell_i(\theta|y_i) = y_{1i} \log(p_{1i}) + (n_{1i} - y_{1i}) \log(1 - p_{1i}) + y_{2i} \log(p_{2i}) + (n_{2i} - y_{2i}) \log(1 - p_{2i}).$$

Solving equations (1) and (2) for p_{1i} and p_{2i} for each study gives,

$$p_{1i}(\theta) = \frac{e^{\alpha_0 + x_i \alpha_1 + b_{1i}}}{1 + e^{\alpha_0 + x_i \alpha_1 + b_{1i}}} \quad (6)$$

$$p_{2i}(\theta) = \frac{e^{\beta_0 + x_i \beta_1 + b_{2i}}}{1 + e^{\beta_0 + x_i \beta_1 + b_{2i}}} \quad (7)$$

Then we plug them into function (5) to calculate the maximum likelihood estimates of the regression parameters. To maximize the function (5) with random effects $b_i = (b_{1i}, b_{2i})^T$, one maximizes

$$\sum_{i=1}^N \int_{b_i} \ell_i(\theta|y_i) f(b_i|\Sigma) db_i, \quad (8)$$

where $f(b_i|\Sigma)$ is a bivariate normal distribution with mean 0 and variance Σ . We use the default method in PROC NLMIXED in SAS for computing this integral, i.e., adaptive Gaussian quadrature as described in Pinheiro and Bates (1995). To find the maximum likelihood estimator we would use Trust Region Optimization in PROC NLMIXED in SAS.[2]

The Wald Chi-square test is commonly used in the generalized linear regression model to test statistical significance of fixed effects. To test the fixed effects $H_0 : L\beta = 0$, where L is the contrast matrix. The Wald Chi-square statistic is given as

$$T^2(\hat{\beta}) = \hat{\beta}^T L^T (L \hat{Y} (\hat{\beta} L^T))^{-1} L \hat{\beta}$$

However, in this case, Wald F test is preferred to handle finite samples and overdispersion.[3] The Wald F statistic is given as

$$F(\hat{\beta}) = T^2(\hat{\beta})/r$$

where r is the number of rows of L matrix and the numerator degrees of freedom. The issue here is to estimate the denominator degrees of freedom (DDF) of the F statistic. In this study, we want to find the DDF approximation methods with the best small sample performance in this meta-analysis of sensitivity and specificity.

We assess the performance of three DDF approximation methods in this study, i.e., normal approximation method, Between-Within DDF, and Kenward-Roger DDF.

First, we assess the performance of the simplest method for DDF approximation, i.e., the normal approximation method. Specifically, we will use DDF = 1000 in the approximate Wald F test to test $H_{01} : \alpha_1 = 0$ and $H_{02} : \beta_1 = 0$ in the joint mixed effect logistic regression model.

Between-Within method (Schluchter and Elashoff, 1990)[4] divided the residual degrees of freedom into the between-subject and within-subject portions. It

checks whether a fixed effect changes within any subject. If so, it assigns within-subject degrees of freedom to the effect; otherwise, it assigns the between-subject degrees of freedom to the effect. In the meta-analysis of sensitivity and specificity, the between-subject degrees of freedom will be applied. We will use DDF = $N - \text{rank}[X]$, where $N = 18$ and $\text{rank}[X] = 2$.

Kenward-Roger method (Kenward and Roger, 1997)[5] present a scaled Wald statistic, together with an F approximation of its sampling distribution, that is to shown to perform well in a range of small sample settings. To test the fixed effect $H_0 : L\beta = 0$, the scaled Wald statistic for mixed model for small samples can be expressed as $F^* = \hat{\beta}^T L^T (L\hat{Y}(\hat{\beta})L^T)^{-1} L\hat{\beta} / \varphi r$, where $\hat{Y}(\hat{\beta})$ is the estimated variance-covariance matrix of $\hat{\beta}$, r is the number of rows of L , and φ is the scale factor. To derive the $F_{r,d}$ approximation to the sampling distribution of F^* , the Satterthwaite method[6] is used by matching the first two moments of F^* with those of $F_{r,d}$, and solving for φ and d :

$$E[F^*] = E[F_{r,d}] = \frac{d}{d-2}.$$

$$\text{Var}[F^*] = \text{Var}[F_{r,d}] = 2\left(\frac{d}{d-2}\right)^2 \frac{r+d-2}{r(d-4)}.$$

The values of d derived is the Kenward-Roger DDF.

However, the Between-Within method and Kenward-Roger method are only used in the non-joint model. If we use them in the joint mixed effect model, the correlation ρ between b_{1i} and b_{2i} is ignored. Therefore, the variance-covariance matrix estimate $\hat{Y}(\hat{\beta})$ would be biased. However, since DDF of 1000 is too aggressive in this model, we expect a smaller inflation in the type I error rate by trying Between-Within DDF approximation and Kenward-Roger DDF approximation comparing to the normal approximation.

3 Data Simulation

To compare the small sample performance of the three DDF approximation methods to test the null hypothesis of fixed effect in the joint mixed effect logistic regression model, we conduct data simulation based on this meta-analysis. For simplicity but without the loss of generality, we hold the true values of $\alpha_1, \beta_1, \alpha_0, \beta_0, \sigma_1^2, \sigma_2^2$ as constant. Specifically, we set $\alpha_1 = 0, \beta_1 = 0, \alpha_0 = 0.05, \beta_0 = 2, \sigma_1^2 = 1, \sigma_2^2 = 1$. We keep the sample size of each study the same as that from the original real data. Specifically, we keep the number of patients with BP elevation n_{1i} and the number of patients without BP elevation n_{2i} the same as real data. Also, the proportion of studies with $x_i = 1$ among all studies is kept the same as that from the real data, which equals $\frac{1}{3}$. Because sensitivity and specificity are inversely proportional, we only take into account ρ with negative values in the joint mixed effect model. To examine the effect of total study number and the value of ρ on the observed type I error rate, we set total study number as 18, 36, 54 for ρ equal to 0, -0.25, -0.5, -0.75, respectively.

In order to generate two correlated binomial responses for each study i as the number of true positives y_{1i} and the number of true negatives y_{2i} , we will draw random numbers from two binomial distributions $B(n_{1i}, p_{1i})$ and $B(n_{2i}, p_{2i})$ respectively, where n_{1i} is the number of patients for whom ABPM shows BP elevation, n_{2i} is the number of patients for whom ABPM shows no BP elevation, p_{1i} is the sensitivity, and p_{2i} is the specificity for study i respectively. To generate correlated p_{1i} and p_{2i} for each study i , first we generate b_{1i} and b_{2i} for each study i from the bivariate normal distribution $N(0, \Sigma)$, then we calculate p_{1i} and p_{2i} based on the link function of the joint mixed effect model.

Table 2: The denominator degrees of freedom (DDF) of Wald F statistic by three different approximation methods in the mixed effect logistic regression model

Methods	Estimated DDF
Normal Approximation	1000
Between-Within Approximation	N-2
Kenward-Roger Approximation	estimated from data

For each scenario, 1000 independent replicates are generated. Then a joint mixed effect logistic regression model is fitted to each dataset to calculate the observed type I error rate using three different DDF approximation methods, which are shown in **Table 2**.

To examine the accuracy of point estimates of the fixed effects in our model, for simplicity without loss of generality, we set all the parameters as follows: $\alpha_0 = 0.05$, $\beta_0 = 2$, $\alpha_1 = 0$, $\beta_1 = 0$, $\sigma_1^2 = 1$, $\sigma_2^2 = 1$, $\rho = 0$. We keep the number of studies and the sample size of each study the same as the original data. Then we generate 1000 independent replicates and fit the joint mixed effect logistic regression model to the each dataset to obtain the average point estimates and standard deviations of α_1 , β_1 , α_0 , β_0 .

Modeling and simulation are conducted by using SAS 9.4 on the Longleaf cluster.

4 Results

The point estimates of α_1 , β_1 , α_0 , β_0 are shown in **Table 3**. Assuming the point estimates follow an approximate normal distribution, all the 95% confidence interval of fixed-effect regression parameters contain the true mean of the population. Therefore, we conclude that the point estimates of fixed-effect regression parameters are unbiased.

Then we compare the small sample performance of three DDF approximation of the Wald F statistic to test null hypothesis of fixed effect used in the joint mixed effect logistic regression model. We examine how the total study number

Table 3: Point estimates of α_1 , β_1 , α_0 , β_0 in the joint mixed model

Parameter	Number of Replicate	Mean	Std.
α_1	1000	-0.0065	0.5294
β_1	1000	-0.0202	0.5143
α_0	1000	0.0268	0.3045
β_0	1000	2.0123	0.2973

and ρ affect the observed type I error rate of three DDF approximation methods, controlling for the sample size of each study, proportions of studies with office BP measurement among all studies.

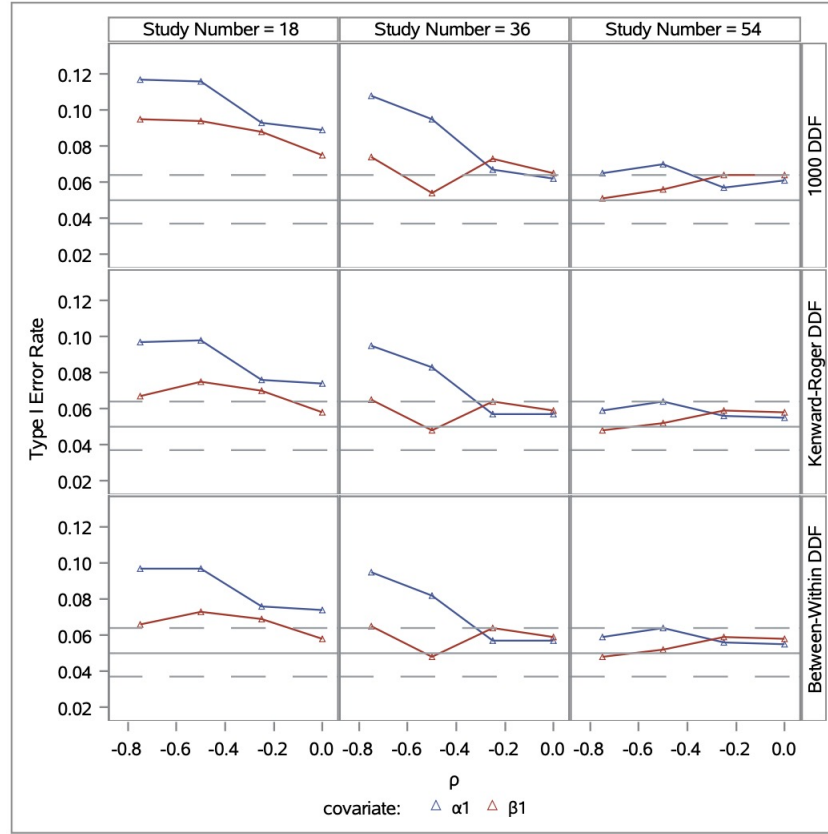


Figure 1: The observed type I error rate of three DDF approximation methods

According to our result, within the range from -0.75 to 0, ρ of larger absolute

value would inflate type I error rate when the total study number is small, i.e., less than 20. When the total study number became larger, ρ has little effect on the type I error rate control of three DDF approximation methods. The effect of ρ is diminished by the increasing total study number. Under our settings, the observed type I error rate fluctuate widely with a total study number of 36 and the fluctuation become smaller when the total study number increases.

The first row of **Figure 1** depicts the observed type I error rate with DDF of 1000, which means we assume the estimated fixed-effect regression parameters follow a normal distribution. As shown in the figure, the observed type I error rate is inflated even if the total study number is larger than 50. Specifically, under our settings, with $\rho = -0.5$ and *studynumber* = 54, the type I error rate of testing $H_0 : \alpha_1 = 0$ is above the 95% confidence interval of the nominal type I error rate.

The observed type I error rate by using Kenward-Roger approximation method and Between-Within approximation method are shown in the second row and the third row, respectively. Under our settings, the estimated DDF by Between-Within method is close to the estimated DDF by Kenward-Roger method, so they share similar type I error control in this study. By using Between-Within method and Kenward-Roger method, the observed type I error rate is less aggressive comparing to normal approximation method. Both of them have a better small sample performance than normal approximation method. As the figure shows, when the study number is smaller than 50, the observed type I error rates under some scenarios are above the 95% confidence interval of the nominal type I error rate. However, under our settings, with a total study number as large as 54, both these two methods maintain the nominal type I error rate. The result of simulations suggest that none of the three methods preserves the type I error rates to nominal level in the joint mixed effect logistic regression model used in the meta-analysis of sensitivity and specificity with a small total number of studies.

We conduct a grid search to find the best denominator degrees of freedom via simulation based on the real data obtained in the meta-analysis. Similarly, we generate 1000 independent replicates setting the parameters as the estimated parameters which are obtained by fitting the joint mixed effect logistic regression model to the real data. Then we conduct a grid search evaluating DDF from 1 to 1000 by calculating the observed type I error rate for each DDF. We found that the best DDF of Wald F statistic to test the null hypothesis $H_0 : \alpha_1 = 0$ that maintains the nominal type I error rate is 4 and the best DDF of Wald F statistic to test the null hypothesis $H_0 : \beta_1 = 0$ that maintains the nominal type I error rate is 9. However, all the three approximation methods would generate a DDF approximation that inflate type I error rate.

5 Conclusion

In this study, we evaluate and compare the small sample type I error rate control of three approximation methods in the joint mixed effect logisitic regression

model used in the meta-analysis of sensitivity and specificity. Our results suggest that the Between-Within method and Kenward-Roger method have a better small sample performance than the normal approximation. Comparing to the normal approximation method, both of them maintain a lower observed type I error rate closer to the 95% CI of the nominal type I error rate when the total study number is small. However, all three methods inflate type I error rate with a small sample size. With large total study number, i.e. as large as 54, all three methods almost maintain the nominal type I error rate.

6 Limitation

We examine how the total study number and ρ affect the observed type I error rate of three DDF approximation methods in the joint mixed effect logistic regression model used in the meta-analysis, controlling for the sample size of each study and the proportions of studies with office BP measurement among all studies. Therefore, the effect of average cluster size, cluster size variation, proportions of studies comparing office BP measurement with ABPM on type I error rate control of three DDF approximation methods need further evaluation. In this study, the average cluster size is 495 with a standard deviation of 366, which might be the cause of the large variance of the observed type I error rate in our study.

None of the three DDF approximation methods find the best DDF that maintains the nominal type I error rate in this study. The approximation method which has an optimal small sample performance in the joint mixed effect model with binary outcomes needs further exploration.

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